LLMFE Public

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preprocessing.py	evaluation script	8 months ago
requirements.txt	added all files	last year
run_llmfe.sh	added all files	last year
utils.py	added all files	last year

About

[TMLR 2026] LLM-FE: Automated Feature Engineering with Large Language Models

- Readme
- MIT license
- Activity
- 90 stars
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- 15 forks

Report repository

Releases

No releases published

Packages

No packages published

Contributors 1

nikhilsab Nikhil Abhyankar

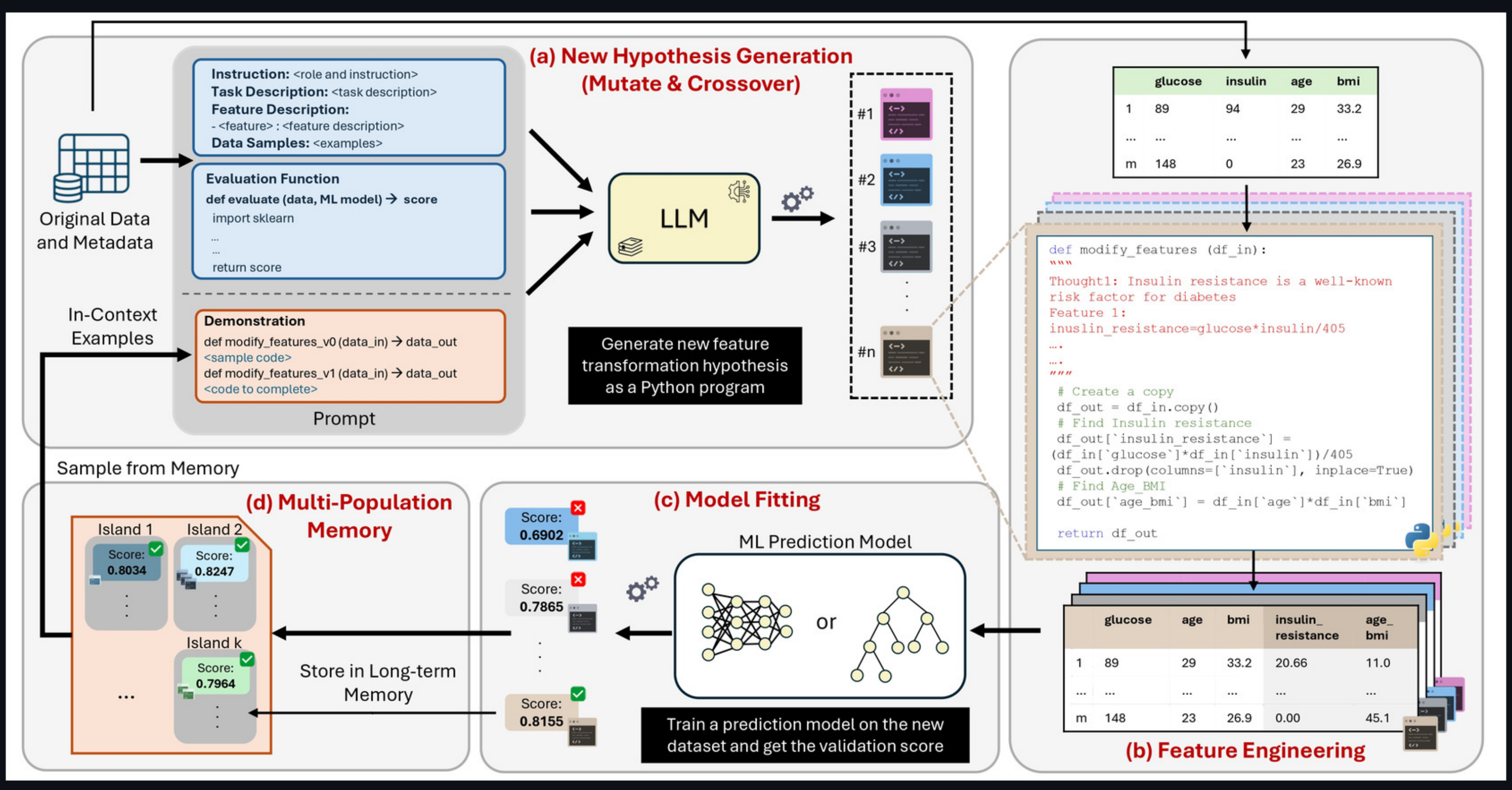
Languages

- **Python** 81.5%
- **Jupyter Notebook** 11.5%
- **Shell** 7%

[TMLR 2026] LLM-FE: Automated Feature Engineering for Tabular Data with LLMs as Evolutionary Optimizers

license MIT · arXiv 2503.14434 · HuggingFace LLMFE

Official implementation of LLM-FE: Automated Feature Engineering for Tabular Data with LLMs as Evolutionary Optimizers.


(a) New Hypothesis Generation (Mutate & Crossover)
Original Data and Metadata
Instruction: <role and instruction>
Task Description: <task description>
Feature Description:
- <feature> : <feature description>
Data Samples: <examples>
Evaluation Function
def evaluate (data, ML model) → score
import sklearn
return score
In-Context Examples
Demonstration
def modify_features_v0 (data_in) → data_out
<sample code>
def modify_features_v1 (data_in) → data_out
<code to complete>
Prompt
LLM
Generate new feature transformation hypothesis as a Python program
#1
#2
#3
#n
Sample from Memory
(d) Multi-Population Memory
Island 1
Score: 0.8034
Island 2
Score: 0.8247
Island k
Score: 0.7964
Store in Long-term Memory
(c) Model Fitting
Score: 0.6902
Score: 0.7865
Score: 0.8155
ML Prediction Model
or
Train a prediction model on the new dataset and get the validation score
(b) Feature Engineering

	glucose	insulin	age	bmi
1	89	94	29	33.2
...	...	...	...	...
m	148	0	23	26.9

```
def modify_features (df_in):
"""
Thought1: Insulin resistance is a well-known
risk factor for diabetes
Feature 1:
inuslin_resistance=glucose*insulin/405
....
....
"""
 # Create a copy
 df_out = df_in.copy()
 # Find Insulin resistance
 df_out[`insulin_resistance`] =
(df_in[`glucose`]*df_in[`insulin`])/405
 df_out.drop(columns=[`insulin`], inplace=True)
 # Find Age_BMI
 df_out[`age_bmi`] = df_in[`age`]*df_in[`bmi`]

 return df_out
```

	glucose	age	bmi	insulin_ resistance	age_ bmi
1	89	29	33.2	20.66	11.0
...	...	...	...	...	...
m	148	23	26.9	0.00	45.1

📄 Overview

LLM-FE is a novel framework that leverages Large Language Models (LLMs) as evolutionary optimizers to automate feature engineering for tabular datasets. LLM-FE iteratively generates and refines features using structured prompts, selecting high-impact transformations based on model performance. This approach enables the discovery of interpretable and high-quality features, enhancing the performance of various machine learning models across diverse classification and regression tasks.

⚙️ Installation

To run the code, create a conda environment and install the dependencies using `requirements.txt`:

```
conda create -n llmfe python=3.11.7
conda activate llmfe
pip install -r requirements.txt
```

🔧 Usage

In `run_llmfe.sh` file, set the OPENAI API key under

```
export API_KEY = <ENTER YOUR API KEY>
```

To run the LLM-FE pipeline on a sample dataset:

```
bash run_llmfe.sh
```

✅ Evaluation

- The **generated features** for each dataset are stored in the logs under the `samples/` folder.
- Install `caafe` into your environment before running the evaluation.

  ```
  conda activate llmfe
  pip install caafe
  ```

- To **run the evaluation**, open the `evaluation.ipynb` notebook and set the `pb_name` variable to the dataset name you want to evaluate (i.e., replace the existing dataset name in `pb_name` with your target dataset), then run the notebook cells.

📝 Citation

```
@article{
abhyankar2026llmfe,
title={{LLM}-{FE}: Automated Feature Engineering for Tabular Data with {LLM}s as Evolutionary (
author={Nikhil Abhyankar and Parshin Shojaee and Chandan K. Reddy},
journal={Transactions on Machine Learning Research},
issn={2835-8856},
year={2026},
url={https://openreview.net/forum?id=qvI35hkpOO}
}
```

📄 License

This repository is licensed under MIT licence.

This work is built on top of other open source projects like FunSearch and LLM-SR. We thank the original contributors of these works for open-sourcing their valuable source codes.

📫 Contact Us

For any questions or issues, you are welcome to open an issue in this repo, or contact us at nikhilsa@vt.edu and parshinshojaee@vt.edu.